In [10]:
```
import numpy as np
import pandas as pd
```

In [11]:
```
data = pd.read_csv("HousingData.csv")
data.head()
```

Out[11]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	ME
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4.98	2
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9.14	2
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4.03	3
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94	3
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	NaN	3

In [12]:
```
data.tail()
```

Out[12]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	M
501	0.06263	0.0	11.93	0.0	0.573	6.593	69.1	2.4786	1	273	21.0	391.99	NaN	
502	0.04527	0.0	11.93	0.0	0.573	6.120	76.7	2.2875	1	273	21.0	396.90	9.08	
503	0.06076	0.0	11.93	0.0	0.573	6.976	91.0	2.1675	1	273	21.0	396.90	5.64	
504	0.10959	0.0	11.93	0.0	0.573	6.794	89.3	2.3889	1	273	21.0	393.45	6.48	
505	0.04741	0.0	11.93	0.0	0.573	6.030	NaN	2.5050	1	273	21.0	396.90	7.88	

In [13]:
```
data.describe()
```

Out[13]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS
count	486.000000	486.000000	486.000000	486.000000	506.000000	506.000000	486.000000	506.000000
mean	3.611874	11.211934	11.083992	0.069959	0.554695	6.284634	68.518519	3.795043
std	8.720192	23.388876	6.835896	0.255340	0.115878	0.702617	27.999513	2.105710
min	0.006320	0.000000	0.460000	0.000000	0.385000	3.561000	2.900000	1.129600
25%	0.081900	0.000000	5.190000	0.000000	0.449000	5.885500	45.175000	2.100175
50%	0.253715	0.000000	9.690000	0.000000	0.538000	6.208500	76.800000	3.207450
75%	3.560263	12.500000	18.100000	0.000000	0.624000	6.623500	93.975000	5.188425
max	88.976200	100.000000	27.740000	1.000000	0.871000	8.780000	100.000000	12.126500

In [14]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 506 entries, 0 to 505
Data columns (total 14 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   CRIM     486 non-null    float64
 1   ZN       486 non-null    float64
 2   INDUS    486 non-null    float64
 3   CHAS     486 non-null    float64
 4   NOX      506 non-null    float64
 5   RM       506 non-null    float64
 6   AGE      486 non-null    float64
 7   DIS      506 non-null    float64
 8   RAD      506 non-null    int64
 9   TAX      506 non-null    int64
 10  PTRATIO  506 non-null    float64
 11  B        506 non-null    float64
 12  LSTAT    486 non-null    float64
 13  MEDV     506 non-null    float64
dtypes: float64(12), int64(2)
memory usage: 55.5 KB
```

In [36]:
```
print("The shape of the data is: ")
data.shape
```

Out[36]:
```
The shape of the data is:
(506, 14)
```

In [16]:
```
data.isnull()
```

Out[16]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MI
0	False	False	False	False	False	False	False	False	False	False	False	False	False	F
1	False	False	False	False	False	False	False	False	False	False	False	False	False	F
2	False	False	False	False	False	False	False	False	False	False	False	False	False	F
3	False	False	False	False	False	False	False	False	False	False	False	False	False	F
4	False	False	False	False	False	False	False	False	False	False	False	False	True	F
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
501	False	False	False	False	False	False	False	False	False	False	False	False	True	F
502	False	False	False	False	False	False	False	False	False	False	False	False	False	F
503	False	False	False	False	False	False	False	False	False	False	False	False	False	F
504	False	False	False	False	False	False	False	False	False	False	False	False	False	F
505	False	False	False	False	False	False	True	False	False	False	False	False	False	F

506 rows × 14 columns

In [17]:
```
data.isnull().sum()
```

Out[17]:
```
CRIM       20
ZN         20
INDUS      20
CHAS       20
NOX         0
RM          0
AGE        20
DIS         0
RAD         0
TAX         0
PTRATIO     0
B           0
LSTAT      20
MEDV        0
dtype: int64
```

In [18]:
```
data.fillna(data.mean(), inplace=True)
```

In [19]:
```
data.isnull().sum()
```

Out[19]:
```
CRIM       0
ZN         0
INDUS      0
CHAS       0
NOX        0
RM         0
AGE        0
DIS        0
RAD        0
TAX        0
PTRATIO    0
B          0
LSTAT      0
MEDV       0
dtype: int64
```

In [29]:
```
X = data.iloc[:,0:13]
Y = data.iloc[:,-1]
```

In [30]:
```
from sklearn.model_selection import train_test_split

X_train,X_test,Y_train,Y_test = train_test_split(X,Y,test_size=0.20,random_state=42)
```

In [31]:
```
print(X_train.shape)
print(X_test.shape)
print(Y_train.shape)
print(Y_test.shape)
```

```
(404, 13)
(102, 13)
(404,)
(102,)
```

In [32]:
```
from sklearn.linear_model import LinearRegression
```

In [33]:
```
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
model = make_pipeline(StandardScaler(with_mean=False), LinearRegression())
model.fit(X_train, Y_train)
```

Out[33]:

```
▸ Pipeline
  ▸ StandardScaler
  ▸ LinearRegression
```

In [34]:

```
model.score(X_test,Y_test)
```

Out[34]:

```
0.6588520195508119
```